

IMPERIAL

MaLD

Knee MRI Movement and Loading Device

Presented by ScanOperators

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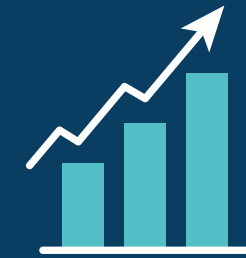


How low can you go :)





Knee Issues in the UK



Did not report
knee pain
53%

Population of
> 50 years old

Reported knee pain
47%

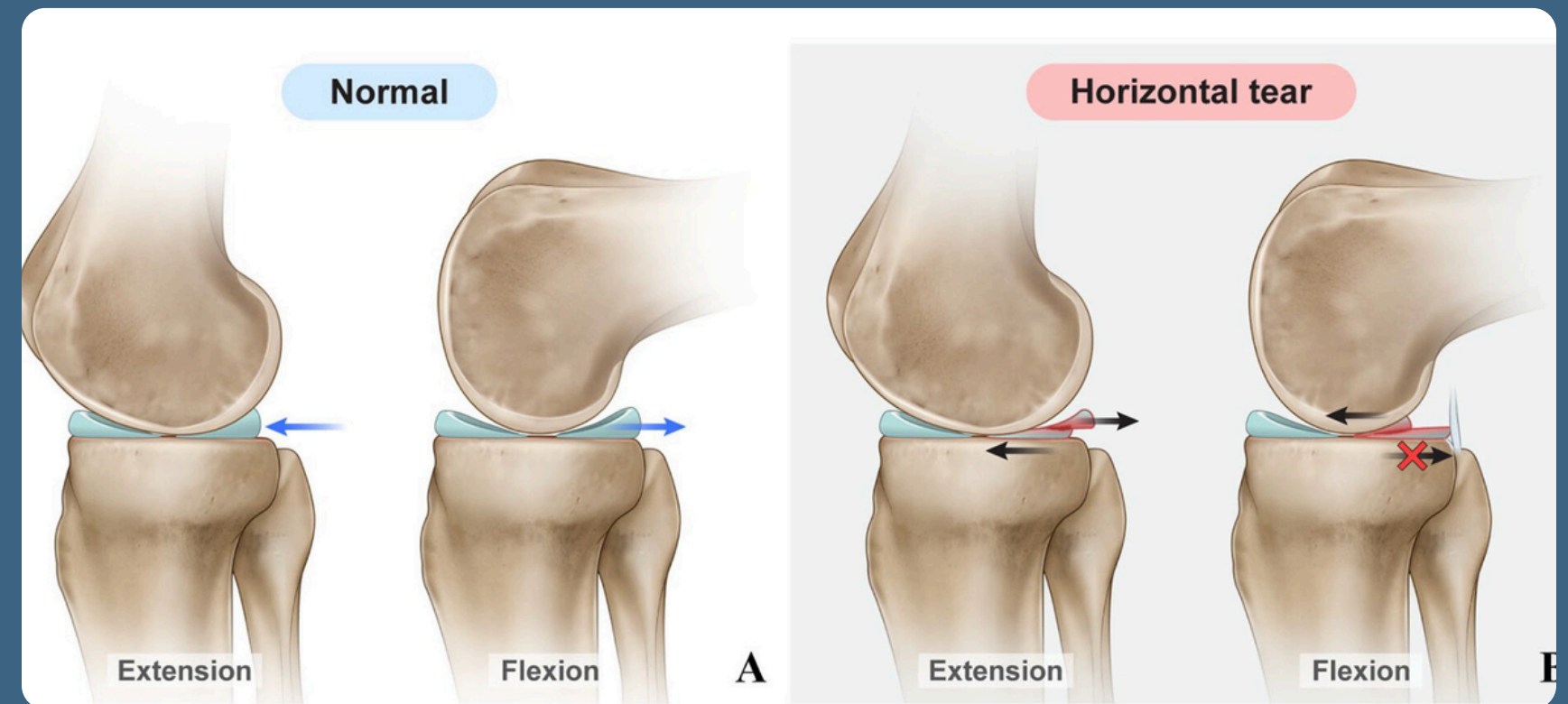
NICE: Jinks et al 2004

Please relax your knees now!



Limitations of Static Imaging

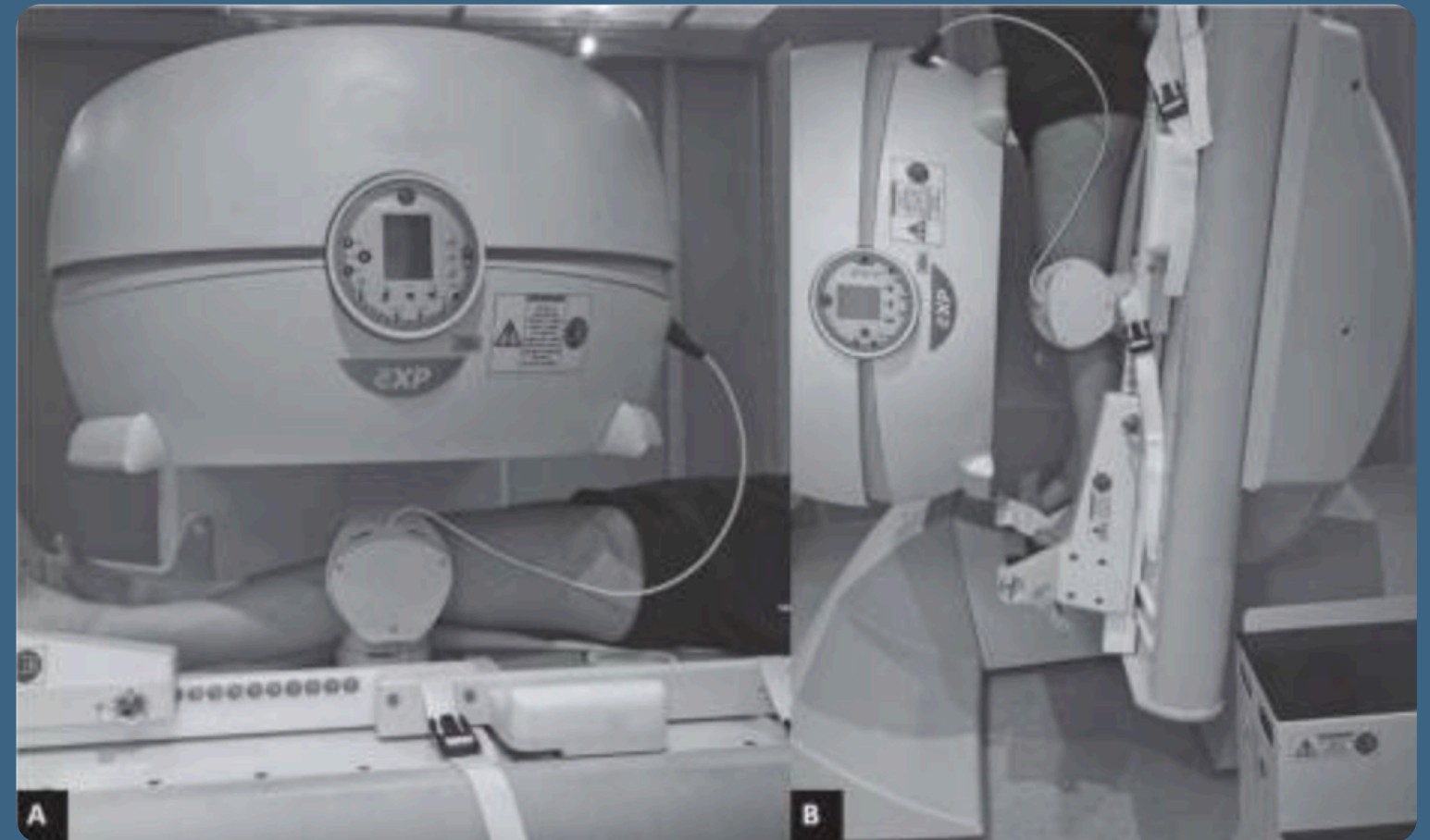
- Cannot capture load-dependent knee mechanics
- **Delayed diagnosis** and treatment



Medial meniscus posterior horn movement in normal and horizontal tear conditions
Adapted from Lim et al., CARTILAGE. 2025

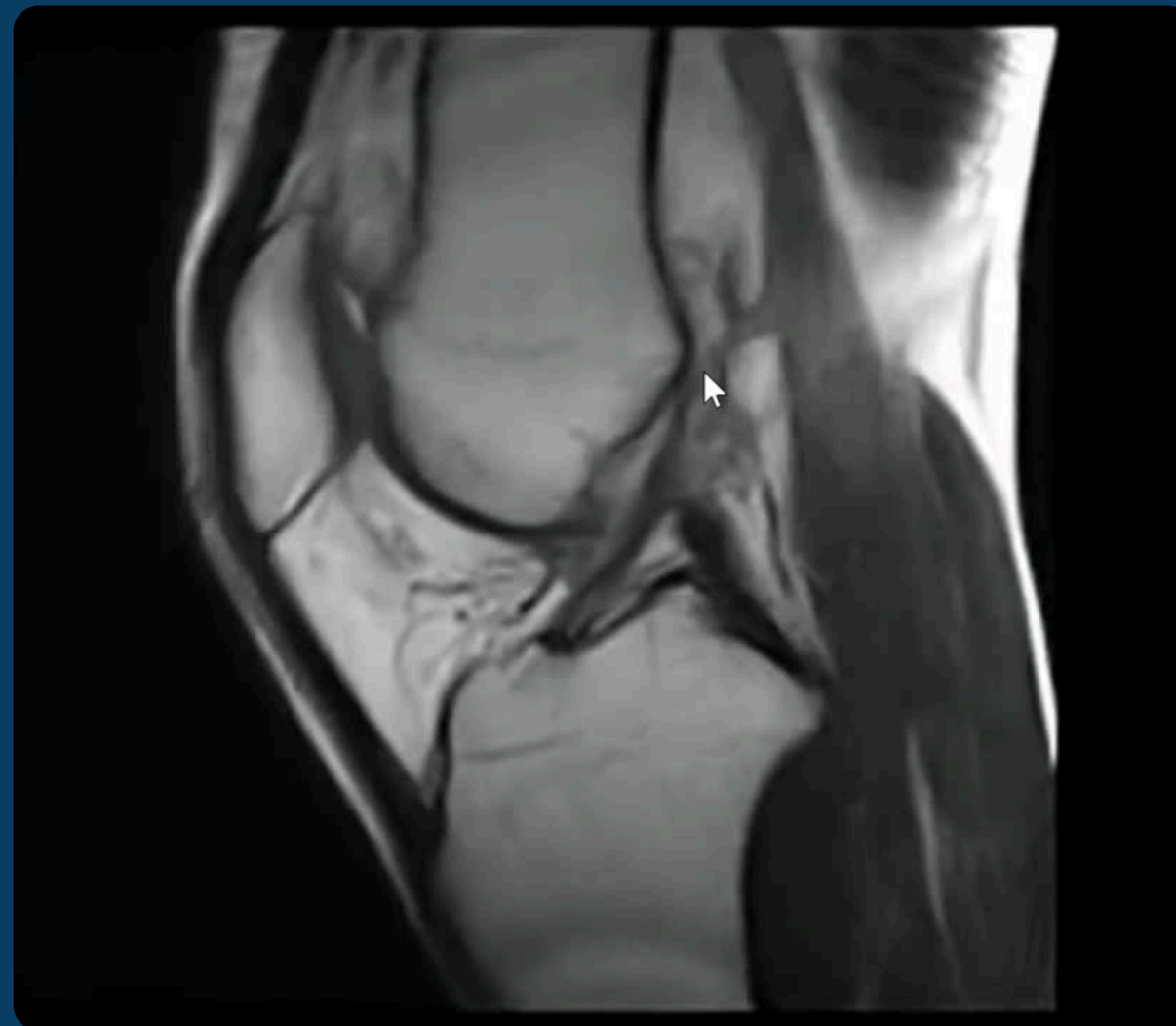
Current Dynamic Solutions

- **Lack repeatability**
- **Too complex and expensive** for wide adoption



WB-MRI weight-bearing MRI Bruno, F et al (2018). Weight-bearing MRI of the knee: a review of advantages and limits.

“Current knee MRIs are not representative of the pathophysiological joint”

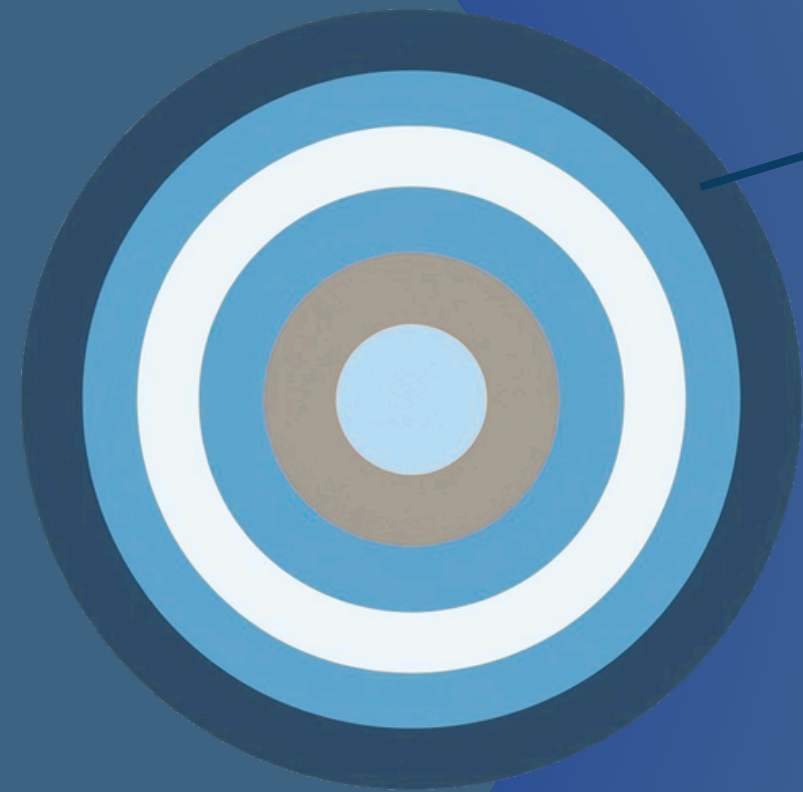


Advanced_MRI. (2020, February 10). Dynamic MRI of knee [Video].
YouTube. <https://www.youtube.com/shorts/Py1L6jhOAnA>

Our Aims ✨



**Simple, accessible and MRI-safe
device**



Our Aims ✨



**Simple, accessible and MRI-safe
device**

Knee loading and positioning

Our Aims ✨



Simple, accessible and MRI-safe device

Knee loading and positioning

Clear clinical indications

Project Schedule



Oct 2025

Clinical scoping
and ideation

Project Schedule



Oct 2025

Clinical scoping
and ideation



Dec 2025

Concept and
design selection

Project Schedule



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Clinical scoping
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Dec 2025

Concept and
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Jan 2026

Detail design and
simulation

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Detail design and
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Feb 2026

Prototype
fabrication,
testing, and Iteration

Design Considerations



Clinical Adaptability

Safe motion and loading limit

Set-up and scan time



Design Considerations



MRI Constraints

Knee coil availability

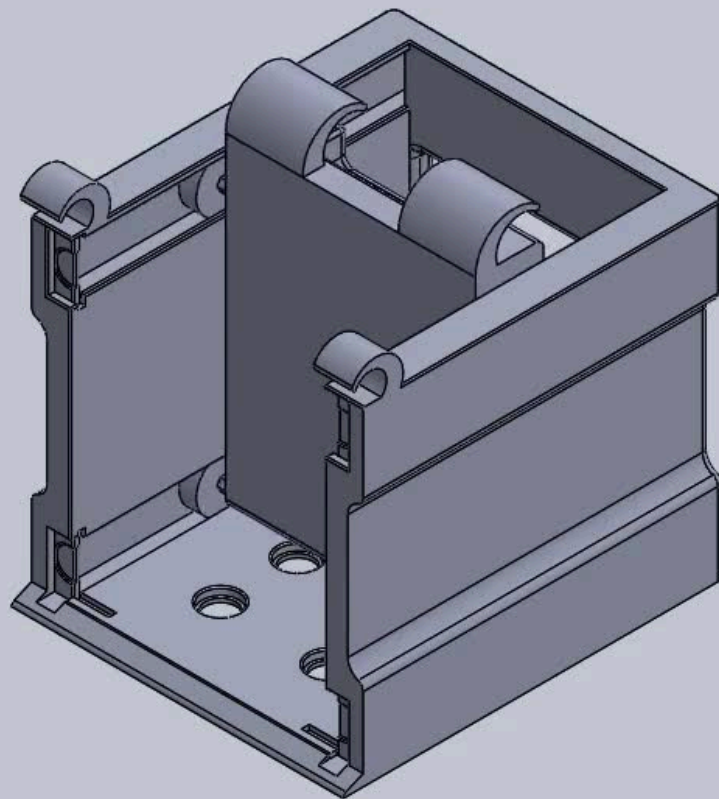
Bore spatial constraints

MRI-safe Technology

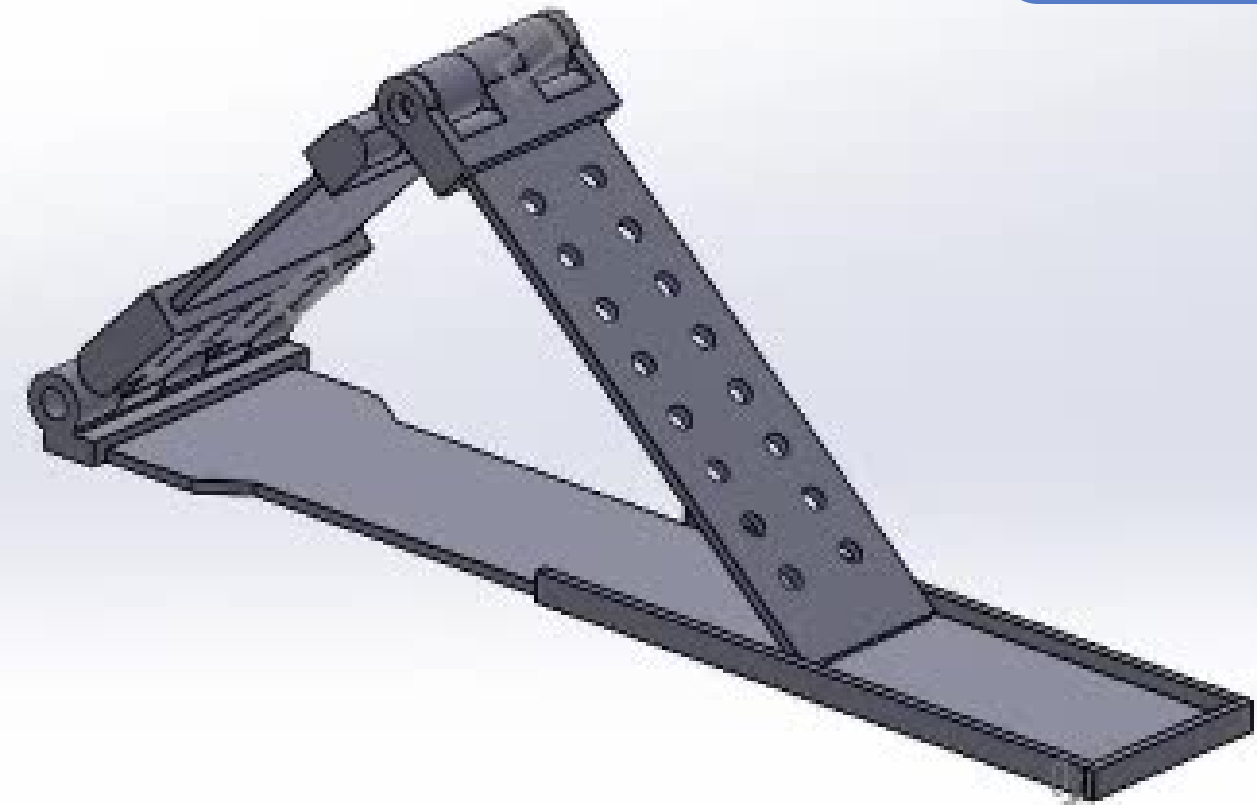


Jacob's Knee

Device Modules



Loading Component



Movement Component

Movement Crew



Amy



Jacob

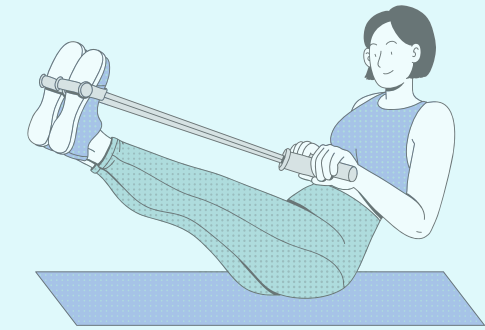


Hannah



Ben

Loading Crew



Sheena



Leo

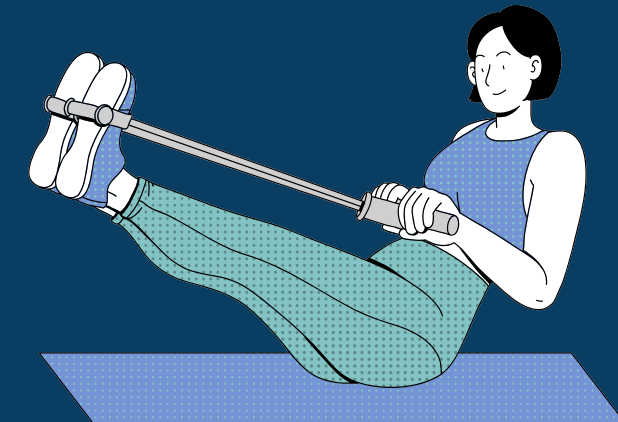


Belle

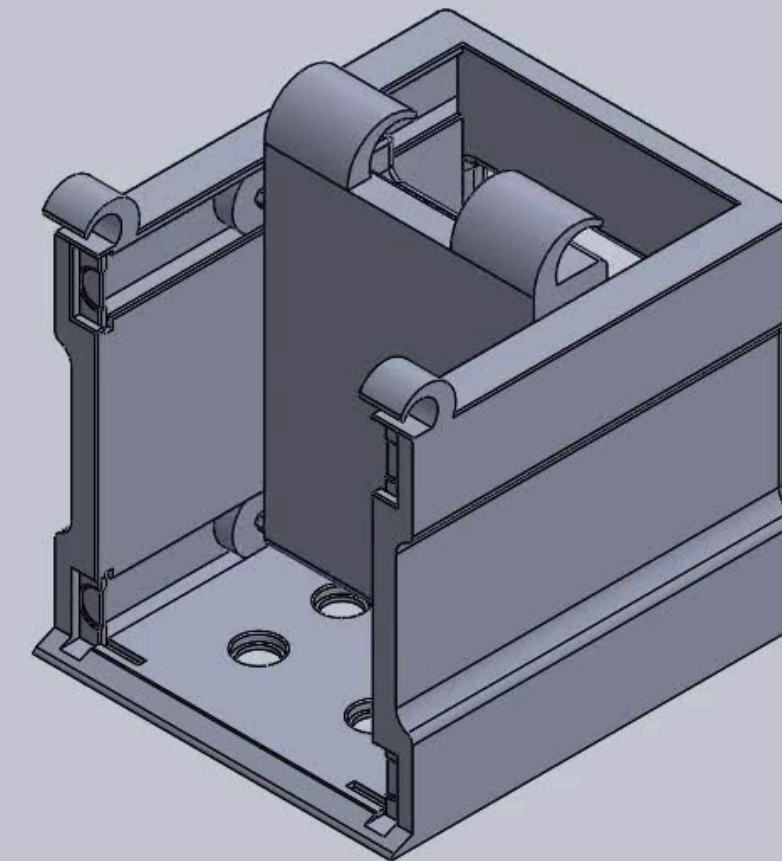


Tayeb

Loading Component



- **Walking-like compression** generated through **resistance bands**
- Push against backplate for **standardising loading**



Movement Component



- **Stable support** at multiple knee angles
- **Patient-controlled rope system**
- **Easy set-up**



Risk Management & Contingency Planning

MRI Incompatibility



Solutions:

- Only **non-ferromagnetic** materials
- Prescan validation checks

Risk Management & Contingency Planning

Patient Discomfort

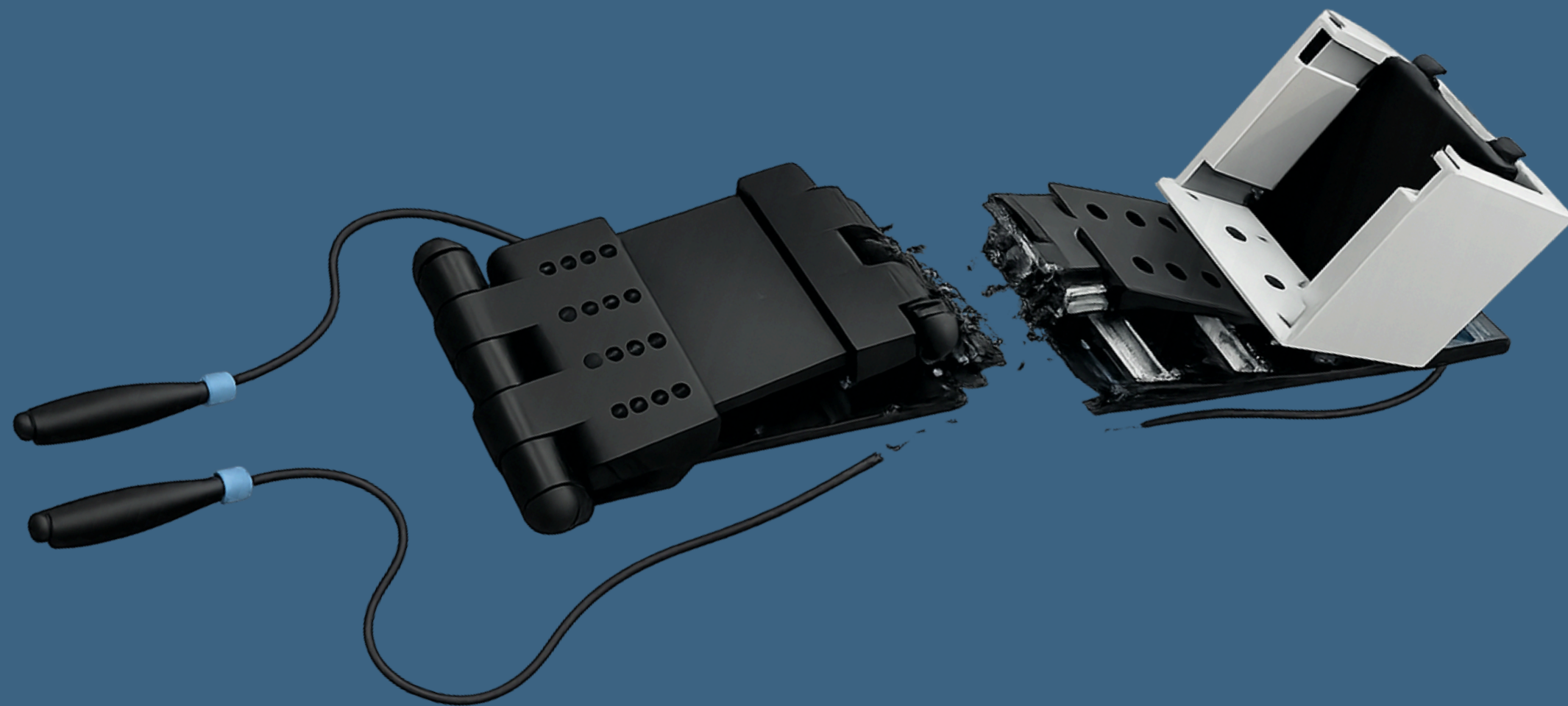


Solutions:

- **Padding**
- Patient tunable load & geometry

Risk Management & Contingency Planning

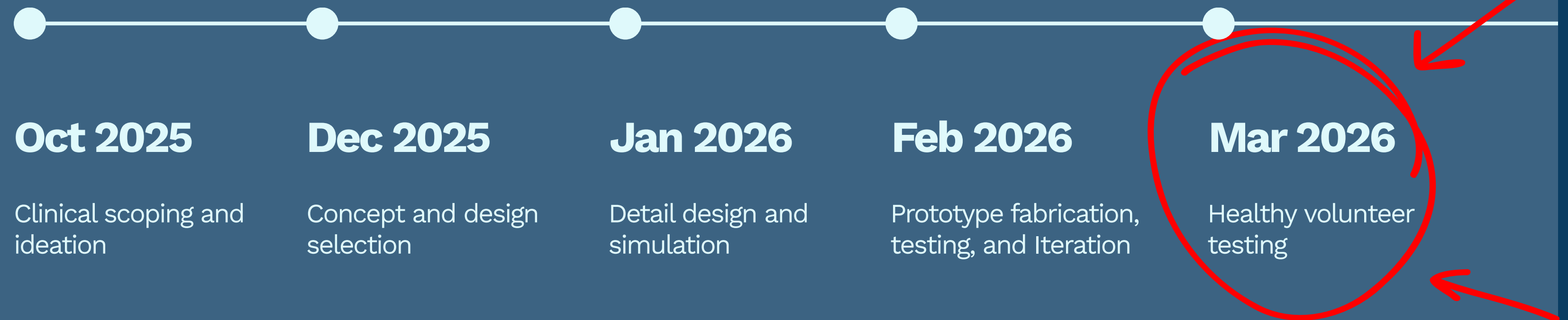
Mechanical Failure

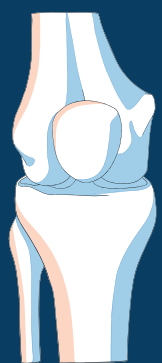


Solutions:

- Replaceable parts
- Over 10 prototypes to be tested

Project Schedule





Healthy Volunteer Testing



Future Directions

Produce radiologist/radiographer training materials



Achieve dynamic loading across multiple angles

Generate functional MRI video of a moving knee



**“We must move away from static
imaging of dynamic joints”**

– NHS Orthopaedic Surgeon